

HICKORY, NC, USA

WMO: 723010

Lat: 35.742N

Lon: 81.382W

Elev: 1175

StdP: 14.08

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 03810

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.3	23.3	-0.2	5.7	28.6	4.1	7.1	29.6	19.9	45.4	18.1	44.7	4.5	320	0.378

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.0	92.0	72.6	89.9	72.4	87.9	71.9	76.2	85.7	75.1	84.2	74.2	82.8	6.2	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.5	130.1	78.8	72.5	125.9	78.2	71.8	122.6	77.7	40.2	86.0	39.3	84.4	38.4	82.7	80.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
17.0	14.7	12.4	DB	12.1	95.7	4.6	3.3	8.8	98.1	6.1	100.0	3.5	101.9	0.2	104.3	(4)
			WB	10.2	77.7	4.2	1.3	7.2	78.7	4.7	79.4	2.4	80.2	-0.7	81.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	59.6	39.8	43.4	50.5	59.7	67.4	74.9	77.4	76.6	70.8	60.1	50.1	42.9
	DBStd	15.12	9.66	8.46	8.95	7.72	6.27	4.54	3.78	3.97	5.60	7.43	7.95	8.27
	HDD50	1035	339	217	108	13	0	0	0	0	0	11	96	251
	HDD65	3419	780	605	455	192	51	2	0	0	17	185	447	685
	CDD50	4523	23	32	122	304	540	748	850	825	623	324	100	32
	CDD65	1432	0	0	4	33	126	300	385	360	190	32	1	1
	CDH74	11572	1	2	49	335	997	2494	3225	2880	1332	242	14	1
	CDH80	4097	0	0	4	55	256	936	1286	1103	417	40	0	0
Wind	WSAvg	4.6	5.1	5.4	5.9	5.9	4.9	4.2	3.8	3.6	3.9	4.0	4.3	4.3
Precipitation	PrecAvg	48.7	3.5	3.9	4.8	3.6	4.4	4.3	4.3	4.0	4.1	3.6	3.4	3.6
	PrecMax	64.1	7.4	7.6	11.8	9.2	8.7	8.5	10.3	10.4	10.9	11.6	7.5	8.3
	PrecMin	33.0	0.4	0.4	1.3	0.1	1.0	0.6	0.1	0.5	0.0	0.0	0.9	0.2
	PrecStd	8.5	1.7	2.0	2.3	2.1	2.0	2.0	2.1	2.2	2.6	2.8	1.7	2.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	68.8	72.2	79.3	85.1	88.9	94.3	94.7	95.5	91.1	84.2	75.9	71.6
		MCWB	56.6	60.6	60.1	64.9	68.8	72.1	74.0	72.8	69.2	69.5	61.0	59.1
	2%	DB	63.7	66.4	74.4	81.4	85.7	90.7	91.6	91.3	88.1	79.9	71.1	65.4
		MCWB	56.2	54.9	58.2	62.8	68.4	71.8	73.9	72.8	69.8	65.5	58.2	58.0
	5%	DB	60.0	63.0	70.3	78.2	82.8	88.3	89.5	88.7	85.0	76.3	67.6	61.1
		MCWB	52.1	53.4	55.9	61.9	67.3	71.5	73.3	73.0	69.8	63.6	57.5	54.5
	10%	DB	55.2	58.5	65.8	74.5	79.9	85.6	87.2	86.1	81.8	72.8	64.1	56.6
		MCWB	48.6	48.8	53.8	60.0	66.3	70.6	72.9	72.2	68.6	62.6	55.8	49.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	62.1	63.1	63.7	68.8	73.0	76.1	77.8	77.4	75.0	73.4	66.7	64.2
		MCDB	65.4	68.5	73.8	78.3	82.1	86.5	88.6	86.8	83.0	78.3	69.0	68.0
	2%	WB	58.5	59.2	61.2	66.6	71.1	74.6	76.4	76.1	73.4	70.1	63.2	60.1
		MCDB	62.3	64.3	69.5	76.4	80.0	85.0	86.5	84.9	80.6	74.8	67.4	63.8
	5%	WB	54.2	55.0	59.1	64.4	69.8	73.4	75.3	75.0	72.2	67.3	60.5	55.6
		MCDB	57.8	60.8	66.8	73.3	78.6	83.6	84.6	83.4	79.2	73.2	65.3	59.4
	10%	WB	49.5	50.5	56.4	62.4	68.4	72.4	74.4	73.9	70.8	64.6	57.2	51.3
		MCDB	54.0	56.2	63.5	70.6	76.3	82.0	83.1	81.9	77.5	70.4	62.2	55.0
Mean Daily Temperature Range	5% DB	MDBR	18.2	19.5	20.3	21.3	20.2	19.1	18.0	18.0	18.2	20.1	20.5	17.5
		MCDBR	23.4	25.6	25.1	25.5	23.7	22.8	21.1	21.5	22.6	23.5	24.9	22.6
		MCWBR	15.6	16.3	13.3	11.6	9.7	7.7	6.9	6.8	7.8	10.9	14.1	15.1
	5% WB	MCDBR	19.1	22.3	20.9	21.3	19.5	19.4	18.3	17.9	17.4	17.6	18.3	18.4
		MCWBR	15.2	16.7	13.3	11.2	8.9	7.6	7.0	6.9	7.4	10.1	13.7	14.3
Clear-Sky Solar Irradiance	taub	0.304	0.313	0.347	0.393	0.442	0.472	0.517	0.509	0.407	0.365	0.320	0.305	
	taud	2.512	2.480	2.425	2.297	2.183	2.136	2.038	2.039	2.310	2.417	2.506	2.539	
	Ebn at Noon	285	295	293	283	269	260	247	246	269	272	277	278	
	Edh at Noon	26	30	34	41	47	49	54	53	38	31	26	24	
All-Sky Solar Radiation	RadAvg	810	1017	1353	1684	1867	1988	1853	1712	1456	1193	902	695	
	RadStd	51	115	91	131	149	142	129	84	133	145	77	66	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+236	+187
Regional (56 neighbors)		+0.92	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+277	+176

Nomenclature: See separate page